Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*AI Agent for Root-Cause Analysis in Manufacturing Systems using Simulation-Generated Operational Logs* *시뮬레이션 기반 제조 운영 로그를 활용한 제조 시스템 문제 원인분석 AI 에이전트 개발*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Analysis Techniques	Software	Software Solutions
Concept of Proposed Technology	**● Goal** Developing an AI-driven Root Cause Analysis (RCA) Agent that interprets comprehensive operational logs (history, master data, dispatching logs) to identify root causes and provide evidence for semiconductor/display manufacturing issues such as Q-time breach, tardy jobs, bottleneck shifts, and lead-time delays. **● Core Functionality** (1) Automation of RCA workflows for given FAB operational problem scenarios: i.e., Semantic routing from user query → Problem identification → Task decomposition → Iterative log tracing & validation for root-cause) (2) Provision of results in a "Cause + Evidence + Analysis history" format. **● Methodology** Adopting high-fidelity Discrete-Event Simulation (DES) and What-if environments to generate synthetic fault-injected scenarios for training and evaluating the RCA Agent, overcoming the risks and costs of real-world testing.			

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	Building multiple agents using large language models with responsibilities such as orchestration, task decomposition, planning, reasoning and evaluation.
Rationale for Joint R&D Project	**1. Strategic Necessity for Semiconductor/Display Autonomous FABs** The semiconductor and display industries are core pillars of South Korea's national strategy. Maintaining global leadership requires an urgent transition toward **autonomous FAB operations**, where operational technology and execution intelligence are as critical as chip/design excellence. Today's FABs are highly complex due to **re-entrant flows** (repetitive operation routes) and movements across multiple areas/facilities (Kim et al., 2021). Furthermore, strict rules like **Q-time** (the maximum allowed waiting time between steps) directly affect product quality and factory stability (Chien et al., 2020). In this complex manufacturing environment, even a small deviation in dispatching, tool availability, or queue conditions can lead to big losses. reducing the time to identify evidence-backed root causes is essential for stable throughput and cycle-time control (Tian et al., 2025)Global best practices are moving beyond rule-based automation toward **agentic AI systems** that can detect abnormalities, diagnose causes, and recommend recovery actions with appropriate operational guardrails (Singla *et al.* 2025; Borsi, 2026). For South Korean companies within the semiconductor/display supply chain, adopting such systems is increasingly a prerequisite for competitiveness in global manufacturing markets. **2. Technical Impasse: Expert-Dependent Root-Cause Analysis of FAB Operation Log** While semiconductor or display fabs are highly digitalized and generate massive volumes of operational logs, the industry faces an **"information paradox"**. Root-cause identification still depends heavily on labor-intensive manual analysis by expert engineers. Effective RCA requires deep understanding of complex constraints (e.g., tool/route/recipe constraints, holds, Q-time, and rework loops) that purely statistical approaches often fail to capture.Recent studies show that **LLM-based agents**—equipped with retrieval and tool-use loops—can address these limitations by dynamically collecting relevant diagnostic information and improving the factuality of Root Cause Analysis (RCA) outputs. (Roy *et al.* 2024.; Patel 2024; Yoshimatsu and Moradi). **3. Data collection via High-Fidelity FAB Simulation** Collecting training data by intentionally inducing abnormal scenarios in a live FAB environment is prohibitively expensive and operationally risky. Therefore, high-fidelity Discrete-Event Simulation (DES) and what-if environments are essential to generate controllable synthetic fault-injection scenarios, producing high-quality datasets for training and benchmark evaluation. Recent academic frameworks emphasize that the accuracy and utility of such synthetic data are the deciding factors in ensuring the reliability of AI models in complex industrial settings. (Chan *et al.* 2022; Buggineni and Camelio 2024). **4. Formalization of Domain Expertise into Diagnostic Workflows** A reliable RCA agent requires converting fragmented expert know-how into structured diagnostic workflows (e.g. problem categorizing → analysis planning → log data tracing → validation). This necessitates a partnership with an organization that possesses causal datasets and process constraints from global top-tier manufacturing sites. By integrating this domain expertise with Explainable AI (XAI), the project will achieve a high-fidelity reasoning engine capable of providing logical evidence for operational problems in complex manufacturing sites.

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	5. Strategic Synergy for Global Market Dominance This joint research leverages a synergistic partnership between a Korean SME with world-class virtual manufacturing DES engines and a premier international research institute such as the University of Toronto specializing in advanced AI. While the SME manages simulation-based FAB operation data generation and domain knowledge structuring, the research institute leads the development of the AI Agent architecture and multi-factor causal reasoning. This collaboration will elevate existing APS solutions into **next-generation autonomous systems**, securing a dominant position in the global smart factory market that neither entity could achieve through standalone efforts.
Key Words	1. Autonomous FAB Operations 2. AI Agent 3. Root-Cause Analysis (RCA) 4. LLM-based Reasoning Agent 5. Manufacturing Operations Log Analytics 6. Discrete-Event Simulation (DES) 7. What-if Simulation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	● An AI Agent for Autonomous Root Cause Analysis (RCA) Engineering an intelligent agent that autonomously deduces the root causes of FAB manufacturing issues (e.g., Q-time breach, bottlenecks, lead-time delays) by analyzing production logs and master data, providing evidence-based reasoning. ● A Simulation-based Train/Validation Pipeline Constructing a high-fidelity synthetic FAB data generation and validation pipeline using Discrete Event Simulation (DES) and What-if analysis to train and evaluate the AI Agent across diverse problem scenarios. ● Advanced Reasoning for Complex/Cascaded Factors Developing a reasoning architecture capable of identifying multiple contributing factors and cascaded causal chains, ensuring scalability for complex global manufacturing sites.			
R&D Target Specifications				

Performance Indicator	Target Specification (Quantitative)	Measuring Method / Validation
Diagnostic Accuracy	Top-1: > 70% (80%), Top-3: > 90%	Knowledge-based experts results vs. Agent inference results
Expert Validity Score	> 3.8 / 5.0	Reasoning logic evaluation by 5+ domain experts.
Problem Coverage	> 3 Types (5 types)	Inclusion of major issue types: tardy jobs, Q-time breach, bottleneck shifts, lead-time delays, or excessive setups.

	Analysis Efficiency	> 70% Reduction (80%)	RCA lead-time compared to manual baseline
	Simulation Scalability	100+ Steps, 10k+ Lots	Capacity for high-complexity models (semiconductor/display scale).

R&D Plan

Period	Foreign Research Institute (UofT)	Korean SME Partner (VMS)
Pre-Period (6 Months) (2026.07 ~2026.12)	● Foundational research on RCA agents ● Evaluation of LLM models ● Architectural framework design	● Acquisition of FAB sample data ● Extraction of production constraints ● Preliminary FAB simulation testing ● Design of scheduling data interfaces ● Q&A Dataset & Documentation of experts' root-cause analyzing process (Easy Level)
Year 1 (12 Months) (2027.01 ~2027.12)	● Installation and pre-training of selected LLM(s) ● Vectorization of knowledge DB ● Performance optimization of Query LLM ● Prototype of the Agent Orchestration model ● Performance validation	● Diversification of FAB sample data ● Formalization of production constraints ● Expansion of scheduling datasets ● Q&A Dataset & Documentation of experts' root-cause analyzing process (Intermediate Level) ● Performance validation
Year 2 (12 Months) (2028.01 ~2028.12)	● Advancement of the Agent Orchestration framework ● Multi-Factor reasoning integration ● Performance validation	● Integration of customer feedback into datasets ● Initial productization (Pilot version) ● Q&A Dataset & Documentation of experts' root-cause analyzing process (Advanced Level) ● Performance validation
Year 3 (6 Months) (2029.01 ~2029.06)	● Final advancement of the Orchestration Agent ● System stabilization and performance tuning	● Final feedback loop integration ● Full productization ● Official market launch ● Collection of real-world user feedback

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

Core Role

- Core AI Agent Development
- Reasoning Architecture Design

Technical Contribution

- AI Agent Architecture Design: Design the AI Agent's reasoning architecture based on FAB manufacturing problem scenarios and RCA (Root Cause Analysis) workflows structured by the SME.
- Agent Implementation & Training: Apply, train, and refine agent performance using high-fidelity simulations and fault-injected scenarios provided by the SME.
- Advanced Reasoning Expansion: Extend the engine to handle multi-factor and cascaded root causes while strengthening Explainable AI (XAI) capabilities.
- Academic validation of the reasoning model's robustness and scalability for complex manufacturing data.

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Expected R&R of Korean SME Partner	**Core Role** • Providing High-fidelity DES (Discrete Event Simulation) and What-if Environments • FAB Sample Model & Data Generation • Productization of Analyzing & Development Framework **Technical Contribution** • Simulation Environment Provision: Provide a high-complexity Discrete Event Simulation (DES) and What-if analysis environment capable of replicating real-world FAB manufacturing failures. • Data Engineering & Dataset Generation: Digitize production planning, scheduling, and operational logs to generate high-quality synthetic FAB datasets for AI training and benchmarking. • Domain Knowledge Structuring: Formalize customer-based FAB expertise (RCA procedures and rules) and establish expert-level validation and labeling standards. • Productization of Analyzing & Validation Framework: Architect and implement a simulation-integrated framework for automated modeling, performance monitoring, AI-based result reasoning, and solution validation. • Practical validation of the Agent through PoC (Proof of Concept) and expert evaluation by global top-tier FAB manufacturing engineers.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Global Competitiveness of the Participating SME** • **Global Market Expansion** Productization of an 'Operational RCA Copilot' that is **deployment-ready** for global FAB manufacturing sites, including the North American market. • **Internalization of Core Technology** Strategic internalization of simulation-based training and validation systems (Knowledge DB + Fault-injected Scenario Generator + DES + What-if Env. + RCA agent) to enhance the global competitiveness of the SME's simulation-driven smart factory solutions. **Operational Efficiency for Semiconductor/Display Manufacturers** • **Operational Productivity Innovation in High-Complexity FABs** Maximizing manufacturing response speed and FAB stability by reducing engineers' root-cause analysis lead time by over 80% through automated AI-driven RCA workflows. Minimizing potential yield loss by enabling rapid response to operational disruptions in high-stakes semiconductor/display production environments. • **Knowledge Digitalization** Digitalizing domain-specific reasoning procedures into structured workflows, ensuring the preservation of critical operational intelligence regardless of personnel turnover. **Systematic Recurrence Prevention** Enable systematic prevention of recurring operational issues, such as Q-time breach,

	bottleneck shifts, and lead-time delays. Establishing a closed-loop diagnostic environment through automated causal reasoning and evidence tracking, providing logical explanations for complex, cascaded manufacturing failures.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization